Guiv Farmanfarmaian

+41 78 900 74 65 guivff@gmail.com

Education

ETH Zürich — *Zürich, Switzerland*

Master's degree in Computer Science — *Sep 2023 – Feb 2026 Expected*

Major: Machine Intelligence

Minor: Theoretical Computer Science

Master's Thesis: *Improving LLM Reasoning on Challenging Problems* (supervised by Dr. Amir Joudaki and Prof. Dr. Thomas Hofmann)

Semester Thesis: *Improving Investment Strategies with GNNs* (supervised by Florian Grötschla, Joel Mathys and Prof. Dr. Roger Wattenhofer; in collaboration with a leading investment firm, under confidentiality)

Bachelor's degree in Mathematics — *Sep 2020 – Sep 2023*

Focus Areas: Number Theory, Theoretical Computer Science

Selected Coursework: Probability Theory, Convex Optimization, Number Theory I & II, APC (Algorithms, Probability and Computing)

Publications

Guiv Farmanfarmaian. *Selection, Recombination, or a Fresh Solve? A Candidate-Free Control for Single-Pass Test-Time Aggregation*. Accepted at the COLM 2026 Workshop on Efficient Reasoning. arXiv preprint, 2026.

Research Interests

- **GRPO Training Dynamics**: Study how reward–KL dependence in group-relative policy optimization shapes training histories: e.g., recurrence versus first discovery of target behaviors under matched marginals and compute
- **AI Safety**: Understand and mitigate failure modes of LLM reasoning — e.g., context-induced answer steering and over-reliance on displayed candidate answers — and design evaluations that isolate them
- **Sample-Efficient Models**: Improve sample efficiency for reasoning, aiming to reach strong performance with substantially less data
- **Distilling Test-Time Improvements into Weights:** Study how gains from test-time compute can be distilled into model weights.
- **Tiny Recursive Models (TRMs) for Algebraic & Symbolic Reasoning**: Extend TRMs beyond ARC-style tasks toward algebraic and symbolic reasoning
- **Continual Learning**: Develop algorithms and architectures that mitigate catastrophic forgetting and enable continual improvement

Research Experience

RL Post-training + Test-time Compute for Math Reasoning (Master's thesis) — ETH Zürich

- **Preventing Mode Collapse in GRPO-based RLVR with teacher-hint conditioning during training (+54% entropy, ~9× lower KL loss compared to GRPO baseline); improved unhinted MATH-500 pass@8 by 3 pp**

- **Recovered RL signal on hard (i.e. zero pass@16) problems with minimal solution prefixes, improving performance without hints on the target benchmark (+5 pp).**
- **Evaluated test-time recursive reasoning and refinement methods on hard math problems**

Supply-chain Graph Learning for Portfolio-relevant Metrics (Semester thesis; ETH × leading investment firm) — ETH Zürich

- Built and evaluated GNN and Graph Transformer models on a large directed supply-chain graph
- Ran ablations and baseline comparisons to test whether graph topology improves financial prediction and portfolio outcomes.

Professional Experience

Autolomous Ltd *London, United Kingdom*
AI Lead *Apr 2026 – Present*

- Lead AI development for autoloMATE, a manufacturing execution system for cell and gene therapy production in GAMP-5 / 21 CFR Part 11 regulated environments
- Build LLM-assisted workflows for regulated documentation, including URS-to-FS requirement mapping and traceability

Funding Circle Ltd *London, United Kingdom*
Data Scientist *Nov 2019 – Aug 2020*

- Implemented regression and clustering methods to analyze portfolio risk
- Automated reporting tools using Python and R
- Implemented data pipelines using AWS
- Participated in company hackathons to develop Random Forest models for risk detection

Reply Sytel (IT Consulting firm) *London, United Kingdom*
Intern *April – May 2017*

- Wrote software (front-end and back-end) to showcase new project initiatives
- Wrote a white paper on the future of networking
- Wrote a white paper on "Proof of Stake", then-novel blockchain technology

Teaching

Teaching Assistant, Computational Intelligence Lab (CIL) *ETH Zürich, Switzerland*
Feb 2025 – Jun 2025

Other

Programming Languages: Python, C++, TypeScript, R, Java
Human Languages: English, French, Persian, Italian (C2-level), German (C2-level)
Citizenship: US, Swiss